

KANAV BHARDWAJ

International Institute of Information Technology, Bangalore

☎ (+91)8837751558

✉ Kanav.Bhardwaj@iiitb.ac.in

🌐 [Kanav Bhardwaj](#)

🔗 [Paradox-73](#)

📁 [Portfolio](#)

Education

International Institute of Information Technology Bangalore (IIIT-B)

Aug 2023 – Jul 2028

Integrated Master of Technology in Computer Science & Engineering

CGPA: 3.26/4.00

Experience

Research Affiliate 🏠 ^{RG} | *IIIT Bangalore*

June 2025 – May 2026

- Developed a predictive autoscaling framework for Dockerized ML workloads utilizing Facebook Prophet and LSTM networks, addressing latency and SLA violations in reactive cloud systems.
- Engineered a simulation pipeline to benchmark 5 resource allocation policies against 14-day synthetic cloud traces featuring diurnal seasonality, stochastic noise, and traffic bursts.
- Demonstrated that a Prophet-driven buffered scaling policy optimized resource allocation, minimizing SLA violations to 8 under-provisioning events while limiting computational cost overhead to 6% above baseline.

Projects

Personal Media Intelligence Hub 🌐 | *Python, XGBoost, Streamlit, SentenceTransformers*

Jan 2026 – June 2026

- Architected a cross-domain Global Taste Engine by unifying 1,250+ media records into a 93-feature semantic space, deploying an interactive Streamlit app to deliver explainable recommendations and multi-modal consumption analytics.
- Mitigated niche-domain sparsity by integrating 7 APIs and compressing textual metadata into low-dimensional vibe components via SentenceTransformers and PCA, while deploying Positive-Unlabeled (PU) learning to model affinity across 3,600+ music tracks.
- Engineered a Ridge-based Stacking Regressor (XGBoost, CatBoost, SVR) utilizing a custom exponential edge-penalty loss to prevent mean-regression, achieving an R^2 of 0.59 in the primary domain and successfully predicting 72.8% of cross-domain items within ± 0.5 stars.

Multi-Pipeline ETL Framework 🌐 | *Python, Hadoop, Pig, Hive, MongoDB, PostgreSQL, Docker*

April 2026 – May 2026

- Built a multi-pipeline ETL framework to process over **3.4M+** semi-structured NASA web server logs, ensuring exact semantic equivalence across MapReduce, Pig, MongoDB, and Hive backends.
- Architected a batch-oriented orchestration pipeline utilizing a Python CLI and PostgreSQL for metadata tracking, facilitating execution traceability and benchmarking across **48** distinct runs.
- Benchmarked four execution paradigms on **1M-record** workloads, identifying MapReduce as the optimal backend (**24.17s** runtime), significantly outperforming Pig (**28.54s**), MongoDB (**32.09s**), and Hive (**93.46s**).

MusicBucket 🌐 🌐 | *React, Supabase, TypeScript, Tailwind CSS*

Mar 2025 – Sept 2025

- Solved a widely requested user problem sourced from Reddit and Spotify forums by architecting a full-stack platform featuring geographical music discovery, custom tier lists, and seamless playlist generation.
- Bypassed strict third-party rate limits for a closed beta of **25** concurrent users by engineering an incremental synchronization pipeline in Supabase to cache initial profiles and exclusively fetch data deltas.
- Adopted Zustand for un-opinionated state management to prevent React context component re-render bottlenecks, eliminating redundant network requests and delivering a zero-latency user experience.

Technical Skills

Languages: Python, C++, Java, TypeScript, JavaScript, LaTeX, SQL

Technologies/Frameworks: React, PyTorch, XGBoost, Scikit-learn, Pandas, Sentence-Transformers, Git, MongoDB

Relevant Coursework

- | | | | |
|--------------------------------|--------------------|------------------------|--------------------------|
| • Data Structures & Algorithms | • Database Systems | • Operating Systems | • NoSQL Systems |
| • Natural Language Processing | • Machine Learning | • Software Engineering | • Recommendation Systems |

Leadership / Extracurricular

8Bit, The Magazine Club of IIITB, Editorial Head

Oct 2023 – Aug 2025

- Led a 16-member team to publish 2 magazine editions, managing content creation, launching a new podcast series, and maintaining the [website](#); also organized 3 large-scale college events (e.g., Treasure Hunt, Pop quiz) for 300+ participants.

Film Divison of IIITB, Club member

Feb 2024 – Present

- Co-hosted technical workshops on scriptwriting and filmmaking for 50+ students and directed **5 winning short film** competition entries (placing 1st, 2nd twice, and 3rd twice).

Achievements

- Awarded the **Best Paper Award** for co-authoring “[Predictive Autoscaling of Dockerised ML Workloads Using Prophet and LSTM](#)” at the 1st International Conference on Recent Advances in Computer Engineering (ICRACE 2026), accepted for publication by Springer Nature / Cureus.
- **JEE 2023 All India Ranks:** Main: **5379** (out of ~ 1.1 M candidates) | Advanced: **7763** (out of ~ 180 K candidates).